

ARBAMINCH UNIVERSITY

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FACULTY OF ELECTRICAL AND COMPUTER ENGINEERING

RESEARCH METHODS AND PRESENTATION PROJECT

Title: Artificial Intelligence System

Instructor: Mr. Tilahun W/C. (MSc)

Prepared by

Name	ID
1.Tesfahun Zeleke	1777/11
2.Liya Markushe	1175/11
3.Michael Tadesse	1296/11
4.Gedeon Eyasu	846/11
5.Wegayehu Semegn	1882/11

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Submitted to: Mr. Tilahun W/C.(MSc)

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CHAPTER ONE

1.1 INTRODUCTION

The term Artificial intelligence refers to a technology that allows for the creation of computers that may exhibit human-like behaviors and actions entirely by artificial methods, independent of any living thing. Robot names are commonly used to refer to artificial intelligence products that, when viewed from an idealist perspective, are entirely human-like and are capable of feeling, perceiving, and making judgments. The capacity of a digital computer or computer-controlled robot to carry out tasks typically performed by intelligent beings, or the simulation of human intelligence in machines that are programmed to think like humans and use computers and machines to mimic the capacity of the human mind for problem-solving and decision-making, The ability to reason and take actions that have the best likelihood of reaching a certain objective is the ideal quality of artificial intelligence. Machine learning (ML), a subtype of artificial intelligence, is the idea that computer programs can automatically learn from and adapt to new data without human assistance. Deep learning algorithms allow for this autonomous learning by ingesting vast quantities of unstructured data, including text, photos, and videos. [1]

1.2 Background History of AI

When we come to History of Artificial Intelligence with Chronological Order is as followes, In May 1. Year, Alexander Heron in antiquity made automatons with mechanical mechanisms working with water and steam power. In 1206, Ebru İz Bin Rezzaz Al Jezeri, one of the pioneers of cybernetic science, has made water-operated automatic controlled machines. In 1623, Wilhelm Schickard invented a mechanic and a calculator capable of four operations. In 1672, Gottfried Leibniz has developed a binary counting system that forms the abstract basis of today's computers. Between year 1822-1859, Charles Babbage developed a mechanical calculator. Ada Lovelace regarded as the first computer programmer because of the work he has done with Babbage's punched cards on his machines. Lovelace's work includes algorithms. In 1923, Karel Capek first introduced the robot concept in the theater play of Rossum's Universal Robots. In 1931, Kurt Gödel introduced the theory of deficiency, which is called by his own name. In 1936, Konrad Zuse developed a programmable computer named Z1 named 64K memory. In 1946, ENIAC (Electronic Numerical Integrator and Computer), the first computer in a room size of 30 tons, started to work.

In 1948, John von Neumann introduced the idea of self replicating program. In 1950, Alan Turing, founder of computer science, introduced the concept of the Turing Test. In 1951, The first artificial intelligence programs for the Mark 1 device were written. In 1956 The logic theorist (Logic Theory-LT) program for solving mathematical problems is introduced by Neweell, Shaw and Simon. The system is regarded as the first artificial intelligence system. By The end of the 1950s, - the beginning of the 1960s A schematic network for machine translation was developed by Margaret Masterman et al. In 1958, John McCarty of MIT created the LISP (list Processing language) language. In 1960, JCR Licklider described the human-machine relationship in his work. In 1962, Unimation was established as the first company to produce robots for the industrial field. In 1965, An artificial intelligence program ELIZA is written. In 1966, The first animated robot "Shakey" was produced at Stanford University. In 1973, DARPA begins development for protocols called TCP / IP. In 1974, The Internet has begun to be used for the first time. In 1978, Herbert Simon earned a Nobel Prize for his limited Rationality Theory, which is an important work on Artificial Intelligence. In 1981, IBM produced the first personal computer. In 1993, Production of Cog, a human-looking robot at MIT, began. In 1997, Deep Blue named supercomputer defeated world famous chess player Kasparov. In 1998, Furby, the first artificial intelligence player, was driven to the market. In 2000, Kismet named robot which can use gesture and mimic movements in communication is introduced. In 2005, Asimo, the closest robot to artificial intelligence and human ability and skill, is introduced. In 2010, Asimo is made to act using mind power. Ben Goertzel and others argued in the early 2000s that AI research had largely given up on the field's original goal of creating artificial general intelligence. Artificial General Intelligence research was founded as a separate sub-field and by 2010 there were academic conferences, laboratories and university courses dedicated to AGI research, as well as private consortiums and new companies. Foundation models, which are large artificial intelligence models trained on vast quantities of unlabeled data that can be adapted to a wide range of downstream tasks, began to be developed in 2018. Models such as GPT-3 released by Open AI in 2020, and Gato released by DeepMind in 2022, have been described as important milestones on the path to artificial general intelligence. [2]

In the above historical lines Artificial intelligence encountered a lot of advancements which enabled it to do many things extraordinarily, some of them includes,

Read - Currently, Summarize Bot which relies on natural language processing, machine learning, artificial intelligence, and block chain technologies, can read whether it's news articles, web links, books, emails, legal documents, audio and image files, and more.

Write - Professional journalists, news organizations such as The New York Times, Washington Post, Reuters, and more rely on artificial intelligence to write. They do, and although this does include many "who, what, where, when, and how" formulaic pieces, AI is able to expand beyond this to more creative writing as well. Many marketers are turning to artificial intelligence to craft their social media posts.

See - Machine vision is when computers can “see” the world, analyze visual data, and make decisions about it. There are so many amazing ways machine vision is used today, including enabling self-driving cars, facial recognition for police work, payment portals, and more. In manufacturing, the ability for machines to see helps in predictive maintenance and product quality control.

Hear and understand - Artificial intelligence is able to detect gunshots, analyze the sound, and then alert relevant agencies. This is one of the mind-blowing things AI can do when it hears and understands sounds.

Speak - Artificial intelligence can also speak. While it’s helpful (and fun) to have Alexa and Google Maps respond to your queries and give you directions, Google Duplex takes it one step further and uses AI to schedule appointments and complete tasks over the phone in very conversational language. It can respond accurately to the responses given by the humans it’s talking to as well.

Smell - There are artificial intelligence researchers who are currently developing AI models that will be able to detect illnesses—just by smelling a human's breath. It can detect chemicals called aldehydes that are associated with human illnesses and stress, including cancer, diabetes, brain injuries, and detecting the "woody, musky odor" emitted from Parkinson's disease even before any other symptoms are identified. Artificially intelligent bots could identify gas leaks or other caustic chemicals, as well. IBM is even using AI to develop new perfumes.

Touch - Using sensors and cameras, there's a robot that can identify "supermarket ripe" raspberries and even pick them and place them into a basket! Its creator says that it will eventually be able to pick one raspberry every 10 seconds for 20 hours a day! The next step for AI tactile development is to link touch with other senses.

Move - Artificial intelligence propels all kinds of movement from autonomous vehicles to drones to robots. The Alter 3 production at Tokyo's New National Theatre features robots that can generate motion autonomously.

Understand emotions - Market research is being aided by AI tools that track a person's emotions as they watch videos. Artificial emotional intelligence can gather data from a person's facial expressions, body language, and more, analyze it against an emotion database to determine what emotion is likely being expressed, and then determine an action based on that info.

Play games - It's not all serious business with artificial intelligence that it can learn to play games such as chess, Go, and poker and, turns out that AI can learn to play these games plus compete and even beat humans at them.

Debate - IBM's Project Debater showed us that artificial intelligence can even be successful at debating humans in complex subjects. Not only is it able to research a topic, but it can also create an engaging point of view and craft rebuttals against a human opponent.

Create - Artificial intelligence can even master creative processes, including making visual art, writing poetry, composing music, and taking photographs. Google's AI was even able to create its own AI "child" that outperformed human-made counterparts.

Read your mind - AI that can read your mind, in that It can interpret brain signals and then create speech. Impressive and life-changing for those with speech impairment, but a little bit unnerving when you consider the mind-reading aspect of the skill. [3]

1.4 Advantage and Disadvantage of AI system

1.4.1 Advantages of AI

AIs main advantages of AI includes:-

Less routine - Artificial intelligence brings about evolution by automating our day-to-day work. Undoubtedly, machines surpass humanity in meticulous tasks. With routine tasks automated, the robots can assume work related to analyses, subtle judgments, and problem-solving. Thus, applying artificial intelligence at work will decrease the workload, empower humans to upgrade their skills. Free from monotonous work, employees will be able to focus on the creative aspects of their jobs. Eventually, this combination of man and machine will make the world a better place.

Less risk - Employing AI capabilities for dangerous tasks will reduce the risk to human well-being and safety. Artificial intelligence technologies will overcome many limitations. For example, it can help to predict wildfire threats and fight it with the help of drones. Aside from that, artificial intelligence takes part in tackling one of the primary causes of climate change - deforestation. AI systems can detect and transmit sounds of illegal logging. Modern AI-driven robotics and autonomous underwater vehicles allow access to the seabed to collect marine ecosystem data and shed new light on the variety of species. AI-powered robots can handle radiation, so they are frequently used in the nuclear energy industry to remove pieces of debris, especially after disasters. Even if they do not eliminate the danger, robots bring great value helping to deal with a catastrophe.

No brakes - To be productive, people need to have refreshment from time to time. A great reinforcement of this human weakness can be obtained with AI-powered machines. They will not get tired, bored, or distracted. Artificial intelligence-based machines will not need a break. Once programmed for long hours, they will work 24x7. Besides, they can take over some extra hard work and with much higher capacity.

No human error - Human beings are prone to physiology, fatigue, stress, emotions, aging, etc. All of these might poorly reflect on decision-making. For example, an overly-stressed doctor or pilot of a falling plane might make a fatal mistake, which will lead to a tragedy, to say nothing of the accidents caused by drunk drivers.

AI computers are error-free (if only adequately programmed). Industries that aim at ensuring a high level of correctness and precision can benefit from adopting artificially intelligent machines. Eliminating costly errors leads these businesses to higher output. Unlike emotional humanity, customized AI can improve the decision-making process, when its set of algorithms uses nothing but previously collected information. Our AI Development Company provide the best expertise in this issue.

No slowdown - Augmentation of human work is also among the pros of artificial intelligence. Capable of running without stopping, AI computing power also bypasses humans in speed and quality of work. It is stronger in modeling and forecasting, combining large data sets. For instance, AI reduces the time of reviewing insurance claims. When it takes days for an insurer to come up with a decision, an artificial intelligence mechanism can cope within minutes. Still, advanced technology needs human insight in shaping the adoption of innovative solutions.

Fewer chores - Apple, Google, and Amazon provide free access to their assistants. Smart technology helps busy people to run their household routine and manage basic tasks. Some AI assistants can even understand the context of a request and synchronize data between all the devices like phone, car, Tv, and even fridge. Thus, it is possible to control them from any point inside and outside the house. AI-powered smart homes not only make life simpler and more dynamic but help to save money and reduce water and energy consumption. High-efficiency systems include appliances that intelligently regulate the use of specific resources. When integrated into a smart network, these devices can reduce costs even more.

Better forecasts - Meteorologists can trace potential severe storms faster by analyzing clouds movements with the help of artificial intelligence. More and more researchers use computer vision and machine learning techniques to detect coming cyclones in real-time. This functionality allows us to properly prepare for natural disasters that impact the lives of millions. Besides, the weather determines the best time to plant and harvest crops. With accurate weather information in hand, farmers will be able to maximize their yield.

Better resources distribution - Power consumed by our houses and machines can be significantly reduced with AI. Whereas, machinery lifetime and efficiency are likely to increase. On a real-time basis, AI can run the most profitable appliance algorithm. For instance, indoor air control improvement relates to AI sensors usage. They adjust and maintain the desired humidity and

temperature in the room. Besides, such a carefully tailored to consumer's preferences approach prevents mechanisms from wasting resources. Not only home utilities are going to benefit from this innovation. Power cars, warehouses, and growing crops equipment will also see reduced energy consumption.

Also, AI can help you to reduce cost, that is highly useful for any kind of business. For instance, you can improve inventory management using AI. Another key opportunity for AI is cargo delivery. AI can find the most effective route, minimizing as many transport distances as possible and saving fuel, in the long term increasing the car lifespan. Other advantages includes Good at detail-oriented jobs, Reduced time for data-heavy tasks, Delivers consistent results and AI-powered virtual agents are always available.

Generally the main advantages advantages of Artificial Intelligence (AI) are:- Finished task faster than a human, Stressful and complex work completed easily, Difficult work done in short period, Various functions can done at a time, Success ratio is high, Less errors in task and defects also, More efficiency in short time, Less space, less size, Calculation of long term and complex situations, and Discover unexplored things. i.e. outer space.

[4] [5]

1.4.2 Disadvantages of AI

A general benefit of AI is based upon cognitive function replication. It has no morals or experience, though. The foremost disadvantages derive from this lack of ethics, and few of them are as follows:-

Fewer workplaces - Will the fourth industrial revolution end up with a robot at your desk? Not really. Still, one way or another, AI will impact more than 95% of jobs either by displacing or complementing them. Moreover, the effect will be uneven for various industries. AI will hit harder laborious areas such as manufacturing, agriculture, and logistics and cause a decline in manual work. Whereas, the healthcare and education spheres are likely to benefit from new technologies.

No abstract thinking - What comes instinctively and without any specific preparation to humans is extremely hard for robots. Artificial intelligence is mindless; "It follows logic algorithms and complex set of instructions. Incapable of comprehensive thinking, artificial intelligence cannot be motivated, inventive, sensitive, or be conscious". For example, philosophy lies outside of the realm of AI scope and is not likely to be conquered by AI.

Lack of ethics - Morality or fairness measurable for a machine is hard to design and convey. AI can hardly be taught what is right unless the engineers provided this concept. The cost-benefit approach will remain the most reliable for robots when they need to estimate a specific situation. To prevent disastrous outcomes, we need to realize the relevance of ethical principles.

More sloth - The convenience technology delivers just one side of the coin. People let AI think on their behalf since there is no need to learn something or memorize. One can always ask a machine. Dependency from a machine is indeed a matter of individual choice. If the technology is used wisely, for delegating and automating simple things to an AI assistant, it will help to grow other parts of the human brain. So, laziness will not be a disadvantage anymore.

Other disadvantages includes implementation is Expensive, Requires deep technical expertise, Limited supply of qualified workers to build AI tools, Only knows what it's been shown; and Lack of ability to generalize from one task to another.

Artificial neural networks and deep learning artificial intelligence technologies are quickly evolving, primarily because AI processes large amounts of data much faster and makes predictions more accurately than humanly possible. While the huge volume of data being created on a daily

basis would bury a human researcher, AI applications that use machine learning can take that data and quickly turn it into actionable information. As of this writing, the main disadvantage of using AI is that it is expensive to process the large amounts of data that AI programming requires.

Generally Some of the main disadvantages of Artificial Intelligence (AI) in our daily lives are as follows, Some time it can be misused leading to mass scale destruction, Programs mismatch sometime done opposite to the command, Human jobs affected, Unemployment problem increased, Creativity is depend upon programmer, Lacks the human touch, Younger generation becomes lazy, Require a lot of time and money, and Technological dependency increased. [4] [5]

1.5 Statement of the problem

Artificial Intelligence (AI) is the toast of every technology driven fields. Adopting and integrating AI technologies is a roller-coaster ride no matter how business-friendly it may sound. A Deloitte report says, around 94% of the AI integrated areas face potential artificial intelligence problems while implementing it. The main problems related to AI includes,

Lack of technical knowledge, To integrate, deploy and implement AI applications in the in any field, the appliers must have the knowledge of the current AI advancement and technologies as well as its shortcomings. The lack of technical know-how is hindering the adoption of this niche domain in most of its application. Only 6% users, currently, have a smooth ride adopting AI technologies.

The **price factor** is another problem where Small and mid-sized organizations struggle a lot when it comes to adopting AI technologies as it is a costly affair. Even big firms like Facebook, Apple, Microsoft, Google, Amazon (FAMGA) allocate a separate budget for adopting and implementing AI technologies. One of the biggest artificial intelligence problems is data acquisition and storage. Business AI systems depend on sensor data as its input. For validation of AI, a mountain of sensor data is collected. Irrelevant and noisy datasets may cause obstruction as they are hard to store and analyze. AI works best when it has a good amount of quality data available to it. The algorithm becomes strong and performs well as the relevant data grows. The AI system fails badly when enough quality data isn't fed into it.

Rare and expensive workforce is another problem as mentioned above, adoption and deployment of AI technologies require specialists like data scientists, data engineers and other SMEs (Subject Matter Experts). These experts are expensive and rare in the current marketplace. Small and medium-sized enterprises fall short of their tight budget to bring in the manpower according to the requirement of the project.

The other is **Issue of responsibility** as the implementation of AI applications comes with great responsibility. Any specific individual must bear the burden of any sort of hardware malfunctions. Earlier, it was relatively easy to determine whether an incident was the result of the actions of a user, developer or manufacturer.

Ethical challenges is also one of the major AI problems that are yet to be tackled are the ethics and morality. The way the developers are technically grooming the AI bots to perfection where it can flawlessly imitate human conversations, making it increasingly tough to spot a difference between a machine and real customer service rep. Artificial intelligence algorithms predict based on the training given to it. The algorithm will label things as per the assumption of data it is trained on. Hence, it will simply ignore the correctness of data, for example- if the algorithm is trained on data that reflects racism or sexism, the result of prediction will mirror back it instead of correcting it automatically.

Lack of computation speed is also an issue since AI, machine learning and deep learning solutions require a high degree of computation speeds offered only by high-end processors. Larger infrastructure requirements and pricing associated with these processors has become a hindrance in their general adoption of the AI technology. In this scenario, cloud computing environment and multiple processors running in parallel offer a potent alternative to cater to these computational requirements. As the volume of data available for processing grows exponentially, the computation speed requirements will grow with it. It is imperative to develop next-gen computational infrastructure solutions.

Illegalness challenges An AI application with an erroneous algorithm and data governance can cause challenges for the company. This is yet again one of the biggest artificial intelligence problems that a developer faces in a real world. Flawed algorithm made with an inappropriate set of data can leave a colossal dent in an organization's profit. An erroneous algorithm will always make incorrect and unfavorable predictions. Problems like data breach can be a consequence of weak & poor data governance–how? To an algorithm, a user's PII (personal identifiable information) acts as a feedstock which may slip into the hands of hackers. Consequently, the organization will fall into the traps of legal challenges.

AI myths and expectations there's quite a discrepancy between the actual potential of the AI system and the expectations of this generation. Media says that artificial intelligence, with its cognitive capabilities, will replace human's jobs. However, the IT industry has a challenge on their hands to address these lofty expectations by accurately conveying that AI is just a tool that can operate only with the indulgence of human brains. AI can definitely boost the outcome of

something that will replace human roles like automation of routine or common work, optimizations of every industrial work, data-driven predictions, etc.

Difficulty of assessing vendors in any emerging field, tech procurement is quite challenging as AI is particularly vulnerable. Businesses face a lot of problems to know how exactly they can use AI effectively as many non-AI companies engage in AI washing, some organizations overstate. It's true that AI technology is a luxurious retreat because you cannot oversee the radical changes it brings into the organization. However, to implement it an organization needs experts who are hard to find. For successful adoption, it needs a high-degree computation processing. Enterprises should concentrate on how they can responsibly mitigate these artificial intelligence problems rather than staying back and ignoring this ground-breaking technology. [6] [7] [8]

1.6 Objectives

1.6.1 General Objective

The main objective of Artificial Intelligence (AI) is to make machines particularly computers do things that require intelligence when done by humans.

1.6.2 Specific Objectives

To have an appreciation for and understanding of both the achievements of AI and the theory underlying those achievements:-

To have an appreciation for the engineering issues underlying the design of AI systems, To have a basic proficiency in a traditional AI language including an ability to write simple to intermediate programs and an ability to understand code written in that language. To have an understanding of the basic issues of knowledge representation and blind and heuristic search, as well as an understanding of other topics such as minimax, resolution, etc. that play an important role in AI programs. To have a basic understanding of some of the more advanced topics of AI such as learning, natural language processing, agents and robotics, expert systems, and planning.

1.7 Methodology

The basic steps used to create an AI from scratch.

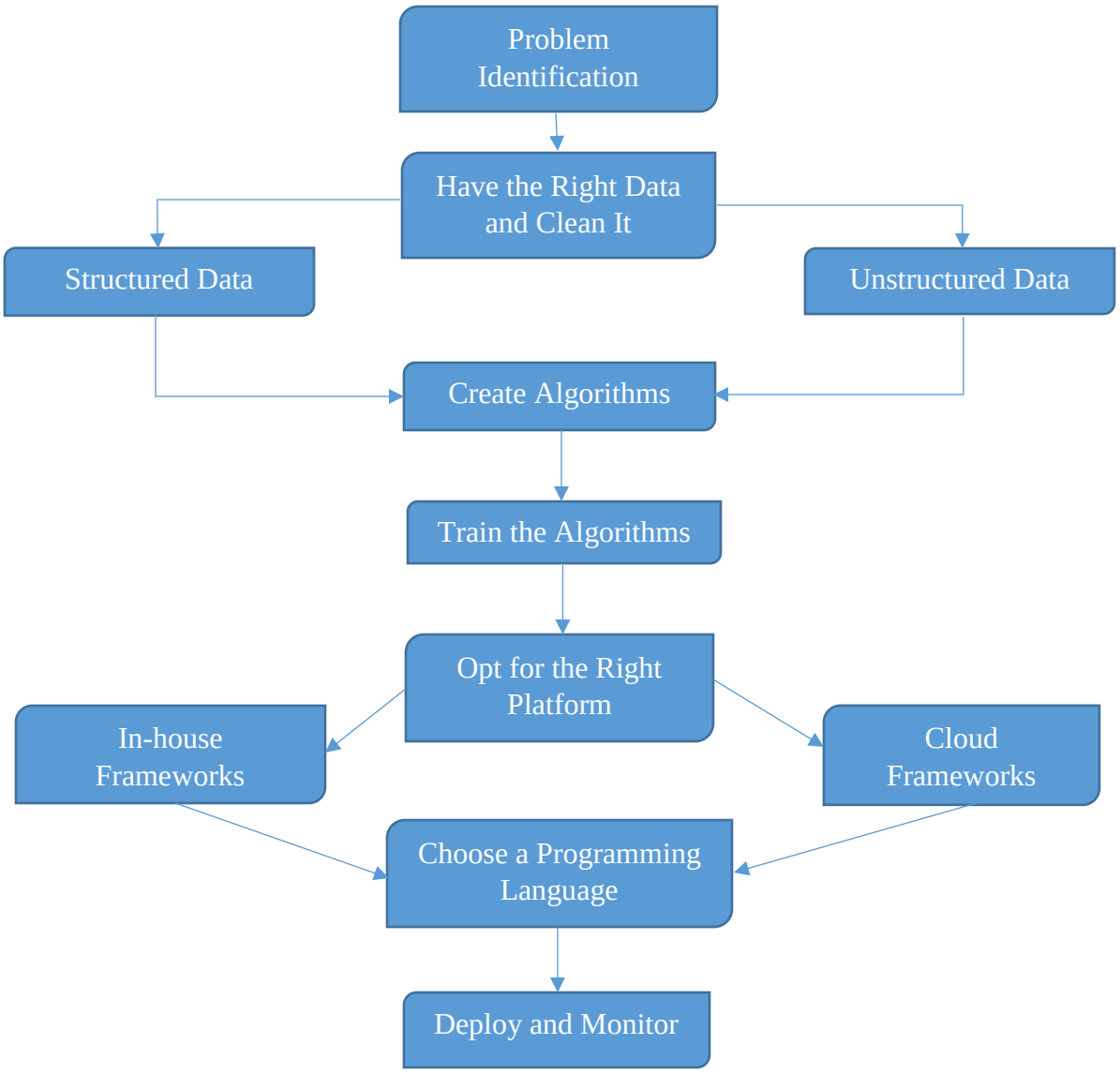


Figure 1Methodology of AI development

In general to make an AI, you need to identify the problem you're trying to solve, collect the right data, create algorithms, train the AI model, choose the right platform, pick a programming language, and, finally, deploy and monitor the operation of your AI system.

1.8 Scope

Real-world applications of AI systems are wide-ranging. Some of the most common scopes of AI in our world includes:-

Speech Recognition also known as automatic speech recognition (ASR), computer speech recognition, or speech-to-text, is a capability that uses NLP to process human speech into a written format. For example, Siri utilizes speech recognition to conduct voice searches.

Customer Service increasingly, more companies are turning to online virtual agents for customer service, thus replacing human agents. According to Savion Global Solutions, 95% of all customer interactions will involve artificial intelligence by 2025.

Computer Vision in this case, AI technology allows computers and systems to derive meaningful information from digital images, videos, and other visual inputs. You can see its application in photo tagging on social media.

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Discovery of Data Trends AI algorithms can use consumers' behavior to discover data trends, allowing companies to build effective cross-selling strategies. As a result, companies can offer relevant add-on recommendations during the checkout process. That's where predictive analytics software steps in. Such software allows real-time decision-making with your data. For instance, the software can generate risk assessment models, such as fraud and risk detection, targeted advertising, and product recommendations.

Fraud Prevention One of the primary problems that artificial intelligence tackles are payment and sensitive information fraud. Companies utilize AI-based systems to detect and prevent this type of fraud effectively.

Automated Stock Trading AI-based high-frequency trading platforms make thousands or, sometimes, millions of trades each day. As of 2020, half of stock market trades in America were automated. According to Allied Market Research, the global algorithmic market size is forecast to account for $31.2 million by 2028.

1.9 Significance

AI is important because, for the first time, traditionally human capabilities can be undertaken in software inexpensively and at scale. AI can be applied to every sector to enable new possibilities and efficiencies.

AI technology is important because it enables human capabilities – understanding, reasoning, planning, communication and perception – to be undertaken by software increasingly effectively, efficiently and at low cost. General analytical tasks, including finding patterns in data, that have been performed by software for many years can also be performed more effectively using AI. The automation of these abilities creates new opportunities in most business sectors and consumer applications. Significant new products, services and capabilities enabled by AI include autonomous vehicles, automated medical diagnosis, voice input for human-computer interaction, intelligent agents, automated data synthesis and enhanced decision-making. AI has numerous, tangible use cases today that are enabling corporate revenue growth and cost savings in existing sectors. Applications will be most numerous in sectors in which a large proportion of time is spent collecting and synthesizing data: financial services, retail and trade, professional services, manufacturing and healthcare. Applications of AI-powered computer vision will be particularly significant in the transport sector. Use cases are proliferating as AI's potential is understood. We describe 31 core use cases across eight sectors: asset management, healthcare, insurance, law & compliance, manufacturing, retail, transport and utilities.

1.10 Organization of the documentation

Organization of the documentation of AI includes to pick the right data sources. It's more critical to get high-quality data than to spend time on improving the AI model itself. Data falls under two categories: structured and unstructured data documentation, Structured data documentation is clearly defined information that includes patterns and easily searchable parameters. For example, names, addresses, birth dates, and phone numbers, whereas Unstructured data doesn't have patterns, consistency, or uniformity. It includes audio, images, infographics, and emails, for better documentation organization next, you need to clean the data, process it, and store the cleaned data before you can use it to train the AI model. Data cleaning or cleansing is about fixing errors and omissions to improve data quality.

CHAPTER TWO

Literatures Reviews

Chandra Kishore, **Raji Atchudan**, **Muthulakshmi Alagan**. July 2022: Explains about Artificial Intelligence (AI) algorithms have shown considerable promise in providing accurate diagnoses with quick data collecting. This work focuses on software models that allow the user to evaluate many different possibilities in the shortest amount of time and is a vital method for proper and dynamic analysis of such entities. The artificial neural network, genetic algorithm, particle swarm optimization, random forest, support vector machine, and extreme learning machine are common AI approaches discussed in this review. This article examines the modern practice and provides recommendations for future machine learning methodologies in fuel cell diagnostic applications

Ishaq Azhar Mohammed September 2020: This paper discussed how artificial intelligence (AI) may be used to address cybersecurity issues. Although secure communication between people, services, and devices is possible via centralized digital organizations, there are significant dangers associated with the digital revolution. Data mining, profiling, and exploitation are can be done without the users' expressed permission. However, the present body of knowledge is deficient in terms of thorough studies on unified communications aspects as well as user privacy and data security measures included in identity management systems. So his paper will address some of how artificial intelligence applies in cybersecurity.

Minh Nguyen, Linh Hoang Truong, Trang T.H. Le July 2021: This paper proposes an efficient algorithm to optimize the transmission and reception of data in UAV-CSS systems, based on the platforms of artificial intelligence (AI) for data processing. The algorithm creates an initial background frame and update to the complete background which is sent to server. It splits the region of interest (moving objects) in the scene and then sends only the changes. This supports the CSS to reduce significantly either data storage or data transmission. In addition, the complexity of the systems could be significantly reduced.

Shrey Bhagat August 2021: Discuses about Machine learning algorithms for structured knowledge, such as the traditional support vector machine and neural network, and therefore the popular deep learning, as well as the tongue process for unstructured knowledge, are typical AI approaches. Cancer, neurology, medical specialties, and strokes are all major disease areas that employ AI technologies.

Gordana Zeba .**Marina Dabic** .Tugrul Daim June2021: This article offers a visual and a comprehensive study of the application of Artificial Intelligence in manufacturing. Existing scholarly literature on Artificial Intelligence in manufacturing, within the Web of Science Core Collection databases, is examined in two periods: 1979-2010 and 2011-2019. These periods are viewed, respectively, as before and after the emergence of the term Industry 4.0. Bibliometric and content analysis of relevant literature is conducted and key findings are subsequently identified. The results indicate that the most important topics today are: cyber-physical systems and smart manufacturing; deep learning and big data; and real-time scheduling algorithms.

Jagadeesh Kengam December 2020: This work describes how Artificial Intelligence can be used and is being used in Educational sector. According to the 21st International Conference on Artificial Intelligence in Education held in 2020, AIED is one of the currently emerging fields in Educational technologies. The use of AI is still unclear for the educators how to make pedagogical advantage of it on a broader scale and how AI can impact on teaching and learning in higher education. The impact of AI in education and its pros and cons are presented here. It also describes a specific way to develop AI enabled platform for education and finally the after effects of AI in education.

Huw Davies, **Cory Salveson**, **Rebecca Eynon** December 2020: In this article, using a relatively novel method for sociology knowledge graph together with Bourdieusean theory, we critically examine how and why different stakeholders in education, educational technology and policy are valorizing AI, the main concepts, such as personalization, they collectively endorse and their incentives for doing so. Drawing on this analysis, we argue that AI is currently being mobilized in education in problematic ways and advocate for more systematic sociological thinking and research to re-orientate the field to account for society's structural conditions.

Lei Wang, **Ang Liu**, **Fei Tao**. **Zhengchao Liu** May 2021: This paper reviews various theories, algorithms, and technologies of AI to different stages of PLM (i.e., product design, manufacturing, and service). A structured roadmap is presented to navigate the future research and application of AI in PLM. This paper also discusses the opportunities and challenges of applying AI for PLM

Bilal Alhayani, Husam Jasim Mohammed, Ibrahim Z. Chaloob March 2021: The aim of the researcher was to determine the effectiveness of artificial intelligence techniques against cyber security risks particularly in case of Iraq, Researcher has opted for quantitative method of research design along with primary data. The researcher collected the data from employees working in this IT industry. The sample size for this study was 468 and confirmatory factor analysis, discriminant validity, basic analysis of model and lastly, hypothesis assessment was carried out. The P-values of all variables were obtained as significant apart from expert system which had no significant relation with artificial intelligence and cyber security. Geographical area, sample size, less variables and accessibility was the main issue.

Iason Gabriel,Sep 2020: This paper looks at philosophical questions that arise in the context of AI alignment. It defends three propositions. First, normative and technical aspects of the AI alignment problem are interrelated, creating space for productive engagement between people working in both domains. Second, it is important to be clear about the goal of alignment. Third, the central challenge for theorists is not to identify 'true' moral principles for AI; rather, it is to identify fair principles for alignment that receive reflective endorsement despite widespread variation in people's moral beliefs. The final part of the paper explores three ways in which fair principles for AI alignment could potentially be identified.

Jose Luis Ruiz Real, et al January 2021,: Artificial Intelligence is a disruptive technology developed during the 20th century, which has undergone an accelerated evolution, underpinning solutions to complex problems in the business world. Neural Networks, Machine Learning, or Deep Learning are concepts currently associated with terms such as digital marketing, decision making, industry 4.0 and business digital transformation. Interest in this technology will increase as the competitive advantages of the use of Artificial Intelligence by economic entities is realised. The aim of this research is to analyse the state-of-the-art research of Artificial Intelligence in business. To this end, a bibliometric analysis has been implement using the Web of Science and Scopus online databases. By using a fractional counting method, this paper identifies 11 clusters and the most frequent terms used in Artificial Intelligence research. The present study identifies the main trends in research on Artificial Intelligence in business and proposes future lines of inquiry.

Jiafu Wan, Xiaomin Li, Hong-Ning Dai, October 2020: This paper focuses on the implementation of AI in customized manufacturing (CM). The architecture of an AI-driven customized smart factory is presented. Details of intelligent manufacturing devices, intelligent information interaction, and construction of a flexible manufacturing line are showcased. The state-of-the-art AI technologies of potential use in CM, i.e., machine learning, multi-agent systems, Internet of Things, big data, and cloud-edge computing are surveyed. The AI-enabled technologies in a customized smart factory are validated with a case study of customized packaging. The experimental results have demonstrated that the AI-assisted CM offers the possibility of higher production flexibility and efficiency. Challenges and solutions related to AI in CM are also discussed.

Jun Liu, Huihong Chang, Jeffrey Yi-Lin Forrest, September 2020: This paper analyzes the impact of artificial intelligence (AI) on technological innovation through logic reasoning and empirical modeling. Based on the industrial robot data provided by the International Federation of Robotics (IFR) and the panel data of China's 14 manufacturing sectors from 2008 to 2017, this paper empirically analyzes the impact of AI on technological innovation. Our analysis shows that the mechanism of how AI affects technological innovation is that the former promotes technological innovation through accelerating knowledge creation and technology spillover, improving learning and absorptive capacities, while increasing R&D and talent investment. The empirical results indicate that under the condition of controlling intensity of R&D investment, FDI, ownership structure, technical imitation, AI significantly promotes technological innovation. And the impact of AI on technological innovation experiences sector heterogeneity: AI has more significant impact on the technological innovation of low-tech sectors. The higher the level of AI, the greater its impact on technological innovation. Based on our established conclusions, we provide corresponding suggestions and recommendations for managerial decision-making.

Yuanbin Wang. **Pai Zheng**, **Pai Zheng**, March 2020: This study provides an introduction of AI techniques. Then, the current development of AI-enabled AM product development is investigated. Existing intelligent agents are used for problems in product design, process design and production stages. Based on the review, current research gaps and future research directions are identified. To guide future development of more efficient and comprehensive intelligent agents, a smart AM framework based on cloud-edge computing is proposed. Global consideration can be realized in the cloud environment, and a fast response can be achieved at the edge nodes.

Khan Muhammad, Salman Khan, Amin Ullah, January 2019: This paper introduces a new paradigm in vision sensor IoT technologies by analyzing the behavior of baby through an intelligent multimodal system. Traditional wearable sensors such as heartbeat if attached to any body part of the baby make him uncomfortable and also some babies are paranoid toward sensors. Our vision based baby monitoring framework employs one of the process improvement techniques known as control charts to analyze the baby behavior.

CHAPTER THREE

3.1 Learning methods AI systems:

Depending on the nature of the "signal" or "feedback" available to the learning system, Machine learning methods are traditionally divided into three broad categories, which correspond to learning paradigms,

Supervised learning: The computer is presented with example inputs and their desired outputs, given by a "teacher", and the goal is to learn a general rule that maps inputs to outputs. **Unsupervised learning:** No labels are given to the learning algorithm, leaving it on its own to find structure in its input. Unsupervised learning can be a goal in itself discovering hidden patterns in data or a means towards an end feature learning.

Reinforcement learning: A computer program interacts with a dynamic environment in which it must perform a certain goal (such as driving a vehicle or playing a game against an opponent). As it navigates its problem space, the program is provided feedback that's analogous to rewards, which it tries to maximize.

Deep learning is of machine learning methods based on artificial neural networks with representation learning. Learning can be supervised, semi-supervised or unsupervised.

Deep reinforcement learning is a subfield of machine learning that combines reinforcement learning (RL) and deep learning. RL considers the problem of a computational agent learning to make decisions by trial and error. Deep RL incorporates deep learning into the solution, allowing agents to make decisions from unstructured input data without manual engineering of the state space.

3.2 skills required for machine learning in AI

To become an AI engineer, you will need to possess the following technical skills:

- Programming Skills - Proficiency in object-oriented programming languages like Python, C#, or C++ is needed to become an AI engineer.
- An understanding of frameworks like Keras and Tensor flow to build AI solutions
- Ability to build deep learning models using neural networks familiarity with CNNs and RNNs
- Familiarity with cloud platforms like Google Cloud, Amazon AWS, or Microsoft Azure you should be able to deploy and scale models using these platforms.
- An understanding of software development methodologies like Agile or Scrum

- Experience with cloud computing
- Data science, data wrangling, and big data knowledge

  In addition to technical techniques above there are other non technical skills required for machine learning in AI which includes:-
- Good communication skills
- The ability to match a business requirement to the end product
- Ability to work in a team

  In addition to the above skills deep knowledge in the following areas is required which includes
- **Statistics** understanding is required in order to learn how these algorithms work. Learn the different types of probability distributions, along with topics like hypothesis testing and sampling, as it helps analyze and visualize data to find unseen patterns. A strong statistical foundation will help to build machine learning models and aid with feature selection and data pre-processing
- **Calculus** is one of the core mathematical concepts in machine learning that permits us to understand the internal workings of different machine learning algorithms. One of the important applications of calculus in machine learning is the gradient descent algorithm, which, in tandem with backpropagation, allows us to train a neural network model.
- **Probability theory** is a mathematical framework for quantifying our uncertainty about the world. It allows us (and our software) to reason effectively in situations where being certain is impossible. Probability theory is at the foundation of many machine learning algorithms.
- **Optimization** is the process of setting decision variable values in such a way that the objective in question is optimized. The optimal solution is a set of decision variables that maximizes or minimizes the objective function while satisfying the constraints.

  Fundamental optimization techniques are typically categorized into *first-order, high-order and derivative-free* optimization includes:-

Gradient descent method is the most popular optimization method. The idea of this method is to update the variables iteratively in the (opposite) direction of the gradients of the objective function. With every update, this method guides the model to find the target and gradually converge to the optimal value of the objective function.

Stochastic gradient descent method was proposed to address the computational complexity involved in each iteration for large scale data. Works by taking the values and adjusting them iteratively based on different parameters in order to reduce the loss function is called back-propagation.

Adaptive Learning rate method decides whether the model will skip certain portions of the data.

Conjugate gradient (CG) approach is used for solving large scale linear systems of equations and nonlinear optimizations problems.

Derivative-Free Optimization Method uses a heuristic algorithm that chooses methods that have already worked well, rather than derives solutions systematically.

Zeroth Order optimization was introduced recently to address the shortcomings of derivative-free optimization to scale to large-size problems and suffer from lack of convergence rate analysis.

Include:

- Ease of implementation with only a small modification of commonly-used algorithms
- Computationally efficient approximations to derivatives when they are difficult to compute
- Comparable convergence rates to first-order algorithms.

Visualization is the practice of translating information into a visual context, such as a map or graph, to make data easier for the human brain to understand and pull insights from. The main goal of data visualization is to make it easier to identify patterns, trends and outliers in large data sets, and it is the representation of data through use of common graphics, such as charts, plots, infographics, and even animations. These visual displays of information communicate complex data relationships and data-driven insights in a way that is easy to understand.

Machine learning algorithm is the method by which the AI system conducts its task, generally predicting output values from given input data, using different types of Neural Networks in Deep Learning which are

- **Artificial Neural Networks (ANN)**

An artificial neural network is an attempt to simulate the network of neurons that make up a human brain so that the computer will be able to learn things and make decisions in a humanlike manner. ANNs are created by programming regular computers to behave as though they are interconnected brain cells. Artificial neural networks are created to digitally mimic the human brain. They are currently used for complex analyses in various fields, ranging from medicine to engineering, and these networks can be used to design the next generation of computers

- **Convolution Neural Networks (CNN)**

A convolutional neural network (CNN) is a type of artificial neural network used in image recognition and processing that is specifically designed to process pixel data, and they are used to help the computer determine features that could be missed in simply flattening an image into its pixel values.

- **Recurrent Neural Networks (RNN)**

Recurrent neural networks (RNN) are the state of the art algorithm for sequential data and are used by Apple's Siri and Google's voice search. It is the first algorithm that remembers its input, due to an internal memory, which makes it perfectly suited for machine learning problems that involve sequential data.

Chapter 4

IMPLEMENTAION OF AI SYSTEMS

In order to implement AI, you need to identify the problem that we are trying to solve, collect the right data, create algorithms, train the AI model, choose the right platform, **pick a programming language**, and, finally, deploy and monitor the operation of your AI system.

From several programming languages, we need to pick the right one that suits the needs of our project. Some of them are,

Python

Python is a computer programming language often used to build websites and software, automate tasks, and conduct data analysis. Python is a general-purpose language, meaning it can be used to create a variety of different programs and isn't specialized for any specific problems.

Python is used in artificial intelligence for

- Building a Neural Network & Make Predictions, Computing the Prediction Error, Understanding How to Reduce the Error, Applying the Chain Rule, Adjusting the Parameters With Backpropagation, Creating the Neural Network Class, Training the Network With More Data, Adding More Layers to the Neural Network.

R

R is a language and environment for statistical computing and graphics. It is a GNU project which is similar to the S language and environment which was developed at Bell.R is widely used in new-style artificial intelligence, involving statistical computations, numerical analysis, the use of Bayesian inference, neural networks and in general Machine Learning. In domains like finance, biology, sociology or medicine it is considered one of the main standard languages.

R uses in AI includes:-

- R is a programming language for statistical computing and graphics that you can use to clean, analyze, and graph your data.
- It is widely used by researchers from diverse disciplines to estimate and display results and by teachers of statistics and research methods.

MATLAB

MATLAB provides AI capabilities similar to those of dedicated AI tools like Caffe and Tensor Flow and more importantly, only MATLAB lets you integrate AI into the complete workflow for developing a fully engineered system. An AI model is just one part of the complete workflow for developing a fully engineered system.

MATLAB makes the hard parts of machine learning easy with: Point-and-click apps for training and comparing models. Advanced signal processing and feature extraction techniques. Automatic machine learning including feature selection, model selection and hyper parameter tuning.

MATLAB provides interactive tools that make it easy to perform a variety of machine learning tasks, including connecting to and importing data. Apps can generate MATLAB code, enabling you to automate tasks. Oftentimes, data has missing or incorrect values.

Octave

Octave is an open-source interactive programming language widely used in machine learning and data analytics courses. Designed primarily for matrix computation, Octave is a powerful tool that enables the development of efficient engineering, data analytics and machine learning programmer.

Octave helps in solving linear and nonlinear problems numerically, and for performing other numerical experiments using a language that is mostly compatible with MATLAB. It may also be used as a batch-oriented language. As part of the GNU Project, it is free software under the terms of the GNU General Public License.

Julia

Julia is a high-level, dynamic programming language, designed to give users the speed of C/C++ while remaining as easy to use as Python. This means that developers can solve problems faster and more effectively. Julia is great for computational complex problems.

Julia allows for dynamic typing: variables don't have types — values have types. Not unlike other high-level languages, these types are determined at runtime. However, it is also possible to assign a type to a variable, just like in static programming.

C++

C++ is used for resource-intensive applications, AI in games and robot locomotion, and rapid execution of projects due to its high level of performance and efficiency. Pros: Fast in-code execution.

C++ has a faster run-time when compared to other programming languages and thus is suitable for machine learning since fast and reliable feedback is essential in machine learning. C++ also has rich library support that is used in machine learning, which we will get to later.

C

It is a programming language for time-sensitive AI/machine learning projects. It works great with statistical AI approach, which is a part of neural networks.

The OS is written in C, the virtual machines are written in C, the compilers are written in C, the interpreters are written in C. Some things are still written in Assembly language, which tends to be even faster. More and more things are being written in something else, which is itself written in C.

IDE used in AI

IDE, or Integrated Development Environment, brings all the different aspects of writing code under a single umbrella – code editor, compiler/interpreter, and debugger. IDEs make it easier to start programming new applications quickly without having to set up different utilities and learn different tools to run a program.

RStudio's primary purpose is to create free and open-source software for data science, scientific research, and technical communication.

RStudio is an integrated development environment (IDE) for R. It includes a console, syntax-highlighting editor that supports direct code execution, as well as tools for plotting, history, debugging and workspace management

PyCharm

PyCharm is a widely used Python IDE created by Jet Brains, This IDE is suitable for professional developers and facilitates the development of large Python projects, Price: Freemium, The most notable features of PyCharm include: Support for JavaScript, CSS, and Type Script, Smart code navigation, Quick and safe code refactoring, Support features like accessing databases directly from the IDE.

IPython/Jupyter Notebook

IPython (Interactive Python) is a command shell for interactive computing in multiple programming languages, originally developed for the Python programming language, that offers introspection, rich me IPython is an interactive shell that is built with python. It provides a more useful shell environment to execute python code in REPL (Read Eval Print Loop). It makes it more interactive by adding features like syntax highlighting, code completion etc. IPython also comes with other tools like Jupyter Notebooks etcdia, shell syntax, tab completion, and history.

Spyder

Spyder is an open-source cross-platform integrated development environment (IDE) for scientific programming in the Python language. Spyder is a very good IDE for compiling the Python code. Spyder executes the code smoothly that creates a very effective debugging skill is a new one also. It can also be use[d] for the data analysis purpose and creating analysis on the basis of data. In our organization, we are also using Spyder for analysis purposes.

Anaconda

Anaconda is a distribution of the Python and R programming languages for scientific computing (data science, machine learning applications, large-scale data processing, predictive analytics, etc.), that aims to simplify package management and deployment.

Rodeo

The open-source, cross-platform Scientific Python Development Environment (Spyder) is an IDE for data science. Spyder is a great option for data scientists because of the IDE's fundamental building elements, which include sophisticated editing, code analysis tools, IPython Console, variable explorer, graphs, debugger, and the help icon.

Google-Colab

Several benefits of Colab over competing services include:

It is unpaid!, Nothing has to be installed on your PC, GPUs (graphics processing units) can be used for demanding computer tasks, You can maintain all of your work in one location because it is integrated with Google Drive.

Platform:

Here is a list of platforms on which AI applications can be deployed:

IBM

Data scientists, developers, and analysts can create, run, and manage AI models, as well as optimize decisions from any location with IBM Watson® Studio. On an open multi cloud architecture, bring teams together, automate AI lifecycles, and accelerate time to value.

Microsoft Azure

Use Azure AI to add speech, vision, language, and decision capabilities to your applications, create chat bots, and uncover insights with AI-powered search. Azure Cognitive Service for Language is a cloud-based service that provides Natural Language Processing (NLP) features for understanding and analyzing text. Use this service to help build intelligent applications using the web-based Language Studio, REST APIs, and client libraries.

Google Cloud

By using google cloud platform You may train models utilizing a variety of customisation options using the AI Platform training service. You can enable distributed training, employ hyper parameter tweaking, and accelerate with GPUs and TPUs while choosing from a wide variety of machine types to power your training jobs. Google Search, predictive analytics, and virtual assistants are examples of narrow AI. The ability for a machine to "detect, think, and act" exactly like a person would be known as artificial general intelligence (AGI).

Amazon

Amazon Machine Learning is an Amazon Web Services product that allows a developer to discover patterns in end-user data through algorithms, construct mathematical models based on these patterns and then create and implement predictive applications.

MyFlow

An open source platform called ML flow is used to control the entire machine learning lifecycle. The following are its main parts: Tracking: enables you to keep track of experiments so you may note and contrast parameters and outcomes.

CHAPTER FIVE

Discussion

AI is at the center of a new enterprise to build computational models of intelligence. The main assumption is that intelligence (human or otherwise) can be represented in terms of symbol structures and symbolic operations which can be programmed in a digital computer.

Conclusion and Recommendation

Conclusion: As can be seen from the discussion above, artificial intelligence technologies make life easier for people and have the potential to give them a competitive advantage in the future. We have covered the definitions of AI, a brief history of AI, applications of AI, advantages and downsides of AI, and many other topics linked to AI during this investigation. Who knows what the AI will be able to achieve for us in the future? Perhaps it will be a robot-dominated society. This is not the end of AI; there is more to come from it.

CHAPETR SIX

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